

Prediction-Powered Data Fusion for Treatment Effect Estimation

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Working draft, August 24, 2026

Randomized controlled trials identify causal effects under transparent design assumptions, but their sample sizes can make average and conditional treatment-effect estimation imprecise. Observational studies are often larger and can yield accurate outcome predictions, but those predictions need not be causal. We develop a model-agnostic prediction-powered framework that keeps the randomized trial as the causal anchor while using observational outcome models and covariates through two tunable channels. The coefficient λ changes the outcome regression inside an RCT-valid pseudo-outcome, and ω introduces a prediction-powered control variate centered across the two samples.

For the average treatment effect, we derive exact conditional unbiasedness, a within-/between-covariate variance decomposition, the joint population variance minimizer, its population-orthogonal simplification, singular and constrained cases, and an honest plug-in algorithm. A separate conditional central limit theorem supports Wald inference after independent coefficient selection, while nested outer cross-fitting requires additional stability analysis for variance estimation. For conditional effects, we formulate a prediction-powered squared-loss learner from an RCT-valid AIPW pseudo-outcome, distinguish pseudo-outcome-noise, fixed-loss-variance, and final learner-risk oracles, and give implementable linear-sieve and frozen-neural-basis versions with honest common-reference selection. We outline an R-loss analogue only as a prospective extension requiring a separate derivation. Exact density-ratio transport covers unequal covariate marginals; classifier odds, Riesz learning, and loss-targeted balancing enter an estimated-weight regime with explicit drift. Existing simulations support the ATE construction in several small-trial regimes. CATE experiments and real-outcome analyses remain planned, so neither the theory nor the current evidence establishes universal finite-sample improvement for estimated coefficients.

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1. Introduction

1.1. Categorized literature review

Prediction-powered inference. Prediction-powered inference uses a large set of machine predictions together with a smaller labeled sample, correcting the prediction-based estimate with labeled residuals so that the statistical target does not require the predictor to be correct (Angelopoulos et al., 2023). This separation between predictive quality and inferential validity is the starting point for the present construction. Randomized observations play the role of causally valid labeled data, while observational outcome models supply potentially useful predictions.

Adaptive prediction-powered inference. PPI++ introduces data-adaptive power tuning that adjusts how strongly predictions enter an estimating equation and reports computational and statistical gains over the original construction (Angelopoulos et al., 2024). Its central lesson is that a prediction should be used in proportion to its empirical value rather than being accepted as truth. Our coefficient ω follows this control-variate logic, but the target and score are causal quantities identified by the randomized design.

Causal data fusion and transport. General causal data-fusion theory distinguishes data collected under different regimes and populations and asks whether a target causal query is identified from their combination (Bareinboim and Pearl, 2016). Reviews of causal data integration likewise separate randomized trials, observational studies, historical controls, and heterogeneous target populations (Shi et al., 2023). The present paper assumes that the trial already identifies the target effect and studies an estimation question: how can observational predictions improve precision without replacing randomized identifying information?

Combining randomized and observational estimates. Existing estimators combine experimental and observational effect estimates through shrinkage, bias detection, or adaptive weighting. Shrinkage estimators can reduce risk when observational bias is sufficiently controlled (Rosenman et al., 2023), while adaptive combination methods move toward the randomized estimator when observational bias is detected (Cheng and Cai, 2021). These approaches combine effect estimates from the two sources. Our construction instead inserts the observational outcome model into an RCT-valid pseudo-outcome and separately centers a prediction-based covariate statistic.

CATE learners and orthogonal losses. Meta-learners convert standard regression algorithms into conditional effect estimators (Künzel et al., 2019). The R-learner residualizes outcomes and treatments to isolate treatment-effect heterogeneity (Nie and Wager, 2021), while doubly robust pseudo-outcome regression can retain favorable error structure under nuisance estimation (Kennedy, 2023). Direct multi-source learners have also been proposed for heterogeneous causal data fusion (Li et al., 2022). These results motivate treating the prediction-powered construction as a loss wrapper that can be paired with a linear sieve or a fixed learned representation.

Covariate shift and balancing. When trial and observational covariate marginals differ, exact density-ratio weighting transports fixed covariate functions between populations. Riesz-representer methods provide an optimization perspective on such linear functionals and on debiasing (Chernozhukov et al., 2021). Density-ratio estimation, generic balancing, and Riesz learning are established tools. The project-specific question is whether they can be targeted to the generated CATE loss class $\{(g - t)^2 : t \in \mathcal{T}\}$ while preserving an RCT-defined risk; we treat this specialization as a working research direction rather than a settled novelty claim.

2. Background and Problem Setting

2.1. Prediction-powered inference and PPI++

Let (X_i, Y_i) , $i = 1, \dots, n$, be labeled pairs and let $\tilde{X}_j$, $j = 1, \dots, N$, be unlabeled covariates. Suppose a fixed predictor supplies $f(X_i)$ and $f(\tilde{X}_j)$. For a mean target, the elementary prediction-powered decomposition is

$$\hat{\mu}_{\text{PPI}} = \frac{1}{N} \sum_{j=1}^N f(\tilde{X}_j) + \frac{1}{n} \sum_{i=1}^n \{Y_i - f(X_i)\}. \quad (1)$$

When labeled and unlabeled covariates share a marginal law and the two samples are independent, the prediction term and residual correction sum to the labeled-population mean for every fixed predictor f . PPI++ generalizes this idea by tuning the strength of the prediction contribution. In our construction, ω is the PPI-style scale on a mean-zero prediction control variate. By contrast, λ blends trial and observational outcome regressions inside an RCT-valid score; it is not the scale on the external prediction mean.

2.2. RCT–OBS problem setting and notation

We observe independent samples

$$D_R = \{V_i^R = (X_i^R, A_i^R, Y_i^R) : i = 1, \dots, n\} \sim P_R^{\otimes n}, \quad (2)$$

$$D_O = \{V_j^O = (X_j^O, A_j^O, Y_j^O) : j = 1, \dots, N\} \sim P_O^{\otimes N}, \quad (3)$$

where X is a covariate vector, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. For a function h , write $\mathbb{P}_{R,n}h = n^{-1} \sum_i h(V_i^R)$ and $\mathbb{P}_{O,N}h = N^{-1} \sum_j h(V_j^O)$.

For the exact adaptive results, these displayed samples are the final-evaluation blocks. Separate RCT and OBS blocks $D_R^{\text{tr}}, D_O^{\text{tr}}$ train nuisance and ratio functions, and separate blocks $D_R^{\text{tu}}, D_O^{\text{tu}}$ select coefficients. Define the pre-evaluation sigma-field

$$\mathcal{F} = \sigma(D_R^{\text{tr}}, D_O^{\text{tr}}, D_R^{\text{tu}}, D_O^{\text{tu}}, \text{prespecified split randomness}). \quad (4)$$

The split assignments are fixed in advance or independently randomized, with their randomness included in $\mathcal{F}$. Conditional on $\mathcal{F}$, every fitted nuisance, ratio, basis, and selected coefficient is fixed and

$$\mathcal{L}(D_R, D_O \mid \mathcal{F}) = P_R^{\otimes n} \otimes P_O^{\otimes N}. \quad (5)$$

For fixed-coefficient statements the tuning blocks may be empty.

Let $Y(1)$ and $Y(0)$ denote potential outcomes. The RCT-population conditional and average treatment effects are

$$\tau(x) = \mathbb{E}_R\{Y(1) - Y(0) \mid X = x\}, \quad \theta = \mathbb{E}_R\{\tau(X)\}. \quad (6)$$

Assumption 1 (Trial identification). *Consistency holds, $\{Y(0), Y(1)\} \perp A \mid X$ under P_R , and the known trial propensity $e(x) = P_R(A = 1 \mid X = x)$ satisfies $\epsilon < e(X) < 1 - \epsilon$ almost surely for a constant $\epsilon > 0$.*

Assumption 2 (Common covariate marginal). *The trial and observational covariates have the same marginal law, $P_R^X = P_O^X$. This assumption does not equate the treatment or outcome mechanisms across sources, and the observational treatment contrast may remain confounded.*

Assumption 3 (Exact covariate transport). *Under covariate shift, $P_R^X \ll P_O^X$ and the exact density ratio*

$$r(x) = \frac{dP_R^X}{dP_O^X}(x), \quad \mathbb{E}_O\{r(X)h(X)\} = \mathbb{E}_R\{h(X)\} \quad (7)$$

is available for every required integrable function h . The common-marginal model is the special case $r = 1$.

Assumption 4 (Honest evaluation and moments). *Every fitted object used in an exact identity is $\mathcal{F}$ -measurable, the final observations satisfy Equation (5), and every displayed conditional moment is finite almost surely. Square-integrability and stronger moment conditions are stated locally when variance estimation or Wald inference requires them.*

The exact-r results use Assumption 3; their common-marginal corollaries use Assumption 2 and set $r = 1$. Neither version assumes that the OBS treatment contrast is causal.

Let $\hat{\mu}_R = (\hat{\mu}_{R,0}, \hat{\mu}_{R,1})$ and $\hat{\mu}_O = (\hat{\mu}_{O,0}, \hat{\mu}_{O,1})$ be outcome regressions fitted on the two sources. Define

$$g(x) = \hat{\mu}_{O,1}(x) - \hat{\mu}_{O,0}(x), \quad (8)$$

$$\varphi(V; q) = q_1(X) - q_0(X) + \frac{A}{e(X)}\{Y - q_1(X)\} - \frac{1 - A}{1 - e(X)}\{Y - q_0(X)\}. \quad (9)$$

Proposition 1 (RCT-validity for an arbitrary outcome regression). *Under Assumptions 1 and 4, every fixed q with finite first moments satisfies*

$$\mathbb{E}_R\{\varphi(V; q) \mid X\} = \tau(X). \quad (10)$$

For $\lambda \in \mathbb{R}$, set

$$\hat{\mu}_\lambda = (1 - \lambda)\hat{\mu}_R + \lambda\hat{\mu}_O, \quad Z_\lambda = \varphi(V; \hat{\mu}_\lambda) = Z_0 + \lambda\Delta, \quad \Delta = Z_1 - Z_0. \quad (11)$$

Equation (10) implies $\mathbb{E}_R(Z_\lambda \mid X) = \tau(X)$ and $\mathbb{E}_R(\Delta \mid X) = 0$. The observational contrast g is used only as a prediction and need not equal τ .

2.3. Existing data-fusion estimators

Three existing strategies provide the nearest comparisons. First, shrinkage methods combine an unbiased experimental estimate with a potentially biased observational estimate and trade variance against observational bias (Rosenman et al., 2023). Second, adaptive RCT–OBS estimators estimate source-specific conditional effects and downweight the observational estimator when its discrepancy from the trial is large (Cheng and Cai, 2021). Third, direct multi-source learners construct weighted pseudo-outcomes for heterogeneous effects across sources (Li et al., 2022). Our estimator differs at the level of the fusion object: λ changes the nuisance regression inside a trial-valid score, while ω centers a fixed prediction across study covariate samples. Neither channel treats the observational treatment contrast as identified causal truth.

Within the adaptive and integrative line, the elastic procedure of Yang et al. (2023) tests whether real-world data are sufficiently compatible with the randomized trial before pooling information for prespecified effect modifiers. The confounding-function approach of Yang et al. (2025) instead introduces and jointly estimates an HTE function and an observational confounding function under explicit cross-source identification conditions. These methods use observational outcomes through a poolability decision or an identified confounding-function model; our two fusion channels retain the RCT-valid score as the causal anchor.

3. Prediction-Powered Fusion for the ATE

3.1. Estimator and exact variance

For fixed (λ, ω) , define

$$\hat{\theta}_r(\lambda, \omega) = \mathbb{P}_{R,n} Z_\lambda + \omega \{ \mathbb{P}_{O,N} \{ r(X)g(X) \} - \mathbb{P}_{R,n} g(X) \}. \quad (12)$$

The first term averages an RCT-valid pseudo-outcome. Under Assumption 3, the expression in braces is a mean-zero control variate; under Assumption 2, this is the special case $r = 1$. Thus, the RCT identifies the target, λ changes within-covariate score noise, and ω uses the large OBS covariate sample to remove predictable between-covariate variation.

Theorem 1 (Exact conditional mean and variance). *Under Assumptions 1, 3, and 4, for positive n, N and square-integrable displayed scores, every fixed (λ, ω) satisfies*

$$\mathbb{E}\{\hat{\theta}_r(\lambda, \omega) \mid \mathcal{F}\} = \theta, \quad (13)$$

$$\text{Var}\{\hat{\theta}_r(\lambda, \omega) \mid \mathcal{F}\} = \frac{1}{n} \text{Var}_R\{Z_\lambda - \omega g(X)\} + \frac{\omega^2}{N} \text{Var}_O\{r(X)g(X)\}. \quad (14)$$

Equivalently, the law of total variance gives the within-/between-covariate decomposition

$$\text{Var}\{\hat{\theta}_r(\lambda, \omega) \mid \mathcal{F}\} = \frac{1}{n} \mathbb{E}_R\{\text{Var}_R(Z_\lambda \mid X)\} + \frac{1}{n} \text{Var}_R\{\tau(X) - \omega g(X)\} + \frac{\omega^2}{N} \text{Var}_O\{r(X)g(X)\}. \quad (15)$$

The mean follows from $\mathbb{E}_R(Z_\lambda) = \theta$ and $\mathbb{E}_O(rg) = \mathbb{E}_R(g)$. The variance follows after rewriting

Equation (12) as the sum of independent RCT and OBS empirical means.

Corollary 1 (Common-marginal specialization). *Under Assumptions 1, 2, and 4, set $r = 1$ in Theorem 1. Then Equation (12) is exactly conditionally unbiased and its OBS variance term is $\omega^2 \text{Var}_R\{g(X)\}/N$.*

3.2. Joint and population-orthogonal oracles

Throughout this subsection, all population moments are conditional on $\mathcal{F}$, the sample sizes n and N are positive, and the displayed random variables have finite second moments. Recall that $\mathcal{F}$ is the sigma-field generated before final evaluation by the prespecified split assignments, the fitted nuisance and ratio functions, and the coefficient choice; conditional on $\mathcal{F}$, all of these objects are fixed while the final RCT and OBS observations retain their evaluation product law.

Define the five conditional population moments

$$A = \text{Var}_R(\Delta), \quad B = \text{Var}_R(g) + \frac{n}{N} \text{Var}_O(rg), \quad (16)$$

$$C = \text{Cov}_R(Z_0, \Delta), \quad D = \text{Cov}_R(Z_0, g), \quad E = \text{Cov}_R(\Delta, g). \quad (17)$$

With

$$\mathbf{H}_{\text{ATE}} = \begin{pmatrix} A & -E \\ -E & B \end{pmatrix}, \quad \boldsymbol{\ell} = \begin{pmatrix} C \\ -D \end{pmatrix}, \quad \beta = \begin{pmatrix} \lambda \\ \omega \end{pmatrix}, \quad (18)$$

the compact variance in Equation (14) becomes

$$\text{Var}\{\hat{\theta}_r(\beta) \mid \mathcal{F}\} = \frac{1}{n} \left\{ \text{Var}_R(Z_0) + 2\boldsymbol{\ell}^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta \right\}. \quad (19)$$

The coefficient vector $\beta = (\lambda, \omega)^\top$ collects the two fusion choices. The vector $\boldsymbol{\ell} = (C, -D)^\top$ is the linear slope of the variance objective at the RCT-only origin $\beta = 0$, while $\mathbf{H}_{\text{ATE}}$ records its quadratic curvature.

The matrix $\mathbf{H}_{\text{ATE}}$ is positive semidefinite because

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H}_{\text{ATE}} \begin{pmatrix} u \\ v \end{pmatrix} = \text{Var}_R(u\Delta - vg) + \frac{n}{N} v^2 \text{Var}_O(rg) \geq 0. \quad (20)$$

Positive semidefiniteness means that the variance objective cannot curve downward along any direction (u, v) . For this two-dimensional matrix, the stronger condition $AB - E^2 > 0$, together with positive semidefiniteness, makes $\mathbf{H}_{\text{ATE}}$ positive definite and invertible. It is therefore the condition that turns a possibly flat quadratic into one with a unique unrestricted minimizer.

Theorem 2 (Joint population oracle). *If $AB - E^2 > 0$, the unique unrestricted minimizer*

of Equation (19) is

$$\lambda_{\text{joint}}^* = \frac{ED - BC}{AB - E^2}, \quad \omega_{\text{joint}}^* = \frac{AD - EC}{AB - E^2}. \quad (21)$$

If $\mathbf{H}_{\text{ATE}}$ is singular, the minimizer set is $-\mathbf{H}_{\text{ATE}}^+ \boldsymbol{\ell} + \text{Null}(\mathbf{H}_{\text{ATE}})$. If $\mathbf{H}_{\text{ATE}}$ is positive definite and the feasible set is nonempty, closed, and convex, the constrained minimizer is unique and equals the $\mathbf{H}_{\text{ATE}}$ -metric projection of the unrestricted minimizer. For a positive-semidefinite $\mathbf{H}_{\text{ATE}}$ and a prespecified compact feasible set, a constrained minimizer exists but need not be unique.

The theorem records the two-dimensional quadratic solution before using the model-implied restriction $E = 0$. Under the exact RCT-valid pseudo-outcome model, the population solution therefore specializes to the axis formulas below; retaining a nonzero interaction remains relevant for the realized empirical tuning quadratic.

Here $\mathbf{H}_{\text{ATE}}$ is singular when $\det(\mathbf{H}_{\text{ATE}}) = AB - E^2 = 0$, in which case the ordinary inverse $\mathbf{H}_{\text{ATE}}^{-1}$ does not exist. Its null space is $\text{Null}(\mathbf{H}_{\text{ATE}}) = \{v \in \mathbb{R}^2 : \mathbf{H}_{\text{ATE}} v = 0\}$, and $\mathbf{H}_{\text{ATE}}^+$ denotes the Moore–Penrose pseudoinverse.

The unrestricted singular formula is well-defined because a variance quadratic is bounded below. This boundedness forces the compatibility condition $\boldsymbol{\ell} \in \text{Range}(\mathbf{H}_{\text{ATE}})$. The term $-\mathbf{H}_{\text{ATE}}^+ \boldsymbol{\ell}$ is one minimum-norm minimizer, and adding any vector in $\text{Null}(\mathbf{H}_{\text{ATE}})$ moves along a direction in which the variance is flat. A nonempty prespecified compact feasible set guarantees that the continuous quadratic attains a constrained minimum even when this minimum is not unique. A prespecified compact feasible set is a nonempty, closed, and bounded subset of $\mathbb{R}^2$ fixed before viewing the tuning outcomes; the implementation below uses the rectangular coefficient box $\mathcal{B}_\beta = [L_\lambda, U_\lambda] \times [L_\omega, U_\omega]$.

Corollary 2 (Joint-oracle variance reduction). *Let $\sigma_0^2 = \text{Var}_R(Z_0)$ and $\mathcal{V}_0 = \sigma_0^2/n$ be the conditional variance at $(\lambda, \omega) = (0, 0)$. If $AB - E^2 > 0$, the unrestricted joint oracle has*

$$V_{\min} = \mathcal{V}_0 - \frac{1}{n} \frac{BC^2 - 2ECD + AD^2}{AB - E^2},$$

$$\mathcal{V}_0 - V_{\min} = \frac{1}{n} \boldsymbol{\ell}^\top \mathbf{H}_{\text{ATE}}^{-1} \boldsymbol{\ell}.$$

The reduction is nonnegative and is strictly positive exactly when $\boldsymbol{\ell} \neq 0$, equivalently when $(C, D) \neq (0, 0)$. Under population orthogonality $E = 0$ and $A, B > 0$, this specializes to

$$V_{\min} = \mathcal{V}_0 - \frac{1}{n} \left(\frac{C^2}{A} + \frac{D^2}{B} \right).$$

If $\mathbf{H}_{\text{ATE}}$ is singular, bounded-below compatibility gives

$$V_{\min} = \mathcal{V}_0 - \frac{1}{n} \boldsymbol{\ell}^\top \mathbf{H}_{\text{ATE}}^+ \boldsymbol{\ell},$$

with a strict reduction exactly when $\boldsymbol{\ell} \neq 0$. For a nonempty compact feasible set $\mathcal{B}_\beta$ containing the origin, the constrained oracle is weakly no worse than $\mathcal{V}_0$; the reduction is strict precisely

when some $\beta \in \mathcal{B}_\beta$ satisfies $2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta < 0$.

For $\rho > 0$, a ridge-stabilized population coefficient is

$$\beta_\rho^* = -(\mathbf{H}_{\text{ATE}} + \rho I_2)^{-1} \ell. \quad (22)$$

It minimizes the penalized quadratic $2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta + \rho \|\beta\|_2^2$. This is a penalized stability oracle, not the oracle for the unpenalized variance in Equation (19). Here I_2 is the 2×2 identity matrix. Adding ρI_2 makes every curvature direction strictly positive, so β_ρ^* is unique even when $\mathbf{H}_{\text{ATE}}$ is singular. The coefficient minimizes the variance quadratic plus the penalty $\rho \|\beta\|_2^2$; it is therefore a penalized oracle rather than another expression for the unpenalized oracle.

The conditional identity $\mathbb{E}_R(\Delta \mid X) = 0$ gives the exact population orthogonality

$$E = \text{Cov}_R\{\Delta, g(X)\} = 0. \quad (23)$$

The identity $\mathbb{E}_R(\Delta \mid X) = 0$ says that changing the outcome-regression blend through λ changes pseudo-outcome noise but not its conditional CATE target. Since $g(X)$ is measurable with respect to X , averaging the conditional mean-zero difference against $g(X)$ yields $E = 0$. Thus the exact population variance has no $\lambda\omega$ interaction, which motivates separate population-axis optimization. It also gives $D = \text{Cov}_R\{\tau(X), g(X)\}$. When $A, B > 0$, the population oracle therefore separates:

$$\lambda^* = -\frac{C}{A}, \quad \omega^* = \frac{D}{B}. \quad (24)$$

If $A = 0$, conditional mean zero forces $\Delta = 0$ almost surely and the λ direction is flat. If $B = 0$, both $g(X)$ and $r(X)g(X)$ are constant on their relevant supports, so the prediction-powered correction is samplewise zero and the ω direction is flat. Under the common covariate marginal,

$$B = \left(1 + \frac{n}{N}\right) \text{Var}_R\{g(X)\}.$$

For the rectangular box $\mathcal{B}_\beta$ and positive A, B , population orthogonality gives the coordinatewise projection

$$\lambda_{\mathcal{B}_\beta}^* = \Pi_{[L_\lambda, U_\lambda]}(-C/A), \quad \omega_{\mathcal{B}_\beta}^* = \Pi_{[L_\omega, U_\omega]}(D/B).$$

For a coupled constraint or a nonzero interaction, the original quadratic must be minimized directly. The population oracle weakly improves the variance objective whenever $(0, 0)$ is feasible. Estimated coefficients have no universal finite-sample no-harm guarantee.

3.3. Estimable honest algorithm

More explicitly, let the tuning-block sizes be $m_R, m_O \geq 2$. For $S \in \{R, O\}$ and $a \in \{1, 2\}$, define

$$\begin{aligned} \bar{h}_{a,S,\text{tu}} &= \frac{1}{m_S} \sum_{k=1}^{m_S} h_a(V_{S,k}), \\ s_{S,\text{tu}}(h_1, h_2) &= \frac{1}{m_S - 1} \sum_{k=1}^{m_S} \{h_1(V_{S,k}) - \bar{h}_{1,S,\text{tu}}\} \{h_2(V_{S,k}) - \bar{h}_{2,S,\text{tu}}\}. \end{aligned}$$

The factor n/N below refers to the final evaluation sizes whose variance is being optimized.

Let $s_R(h_1, h_2)$ and $s_O(h_1, h_2)$ denote unbiased sample covariances on independent tuning blocks, with $s_S^2(h) = s_S(h, h)$. Compute

$$\hat{A} = s_R^2(\Delta), \quad \hat{B} = s_R^2(g) + \frac{n}{N} s_O^2(rg), \quad (25)$$

$$\hat{C} = s_R(Z_0, \Delta), \quad \hat{D} = s_R(Z_0, g), \quad \hat{E} = s_R(\Delta, g). \quad (26)$$

Define

$$\hat{\mathbf{H}}_{\text{ATE}} = \begin{pmatrix} \hat{A} & -\hat{E} \\ -\hat{E} & \hat{B} \end{pmatrix}, \quad \hat{\ell} = \begin{pmatrix} \hat{C} \\ -\hat{D} \end{pmatrix}, \quad \hat{d} = \hat{A}\hat{B} - \hat{E}^2.$$

Choose positive thresholds t_A, t_B, t_{det} before observing the tuning outcomes. The population-orthogonality axis rule is

$$\hat{\lambda}_{\text{axis}} = \begin{cases} -\hat{C}/\hat{A}, & \hat{A} > t_A, \\ 0, & \hat{A} \leq t_A, \end{cases} \quad \hat{\omega}_{\text{axis}} = \begin{cases} \hat{D}/\hat{B}, & \hat{B} > t_B, \\ 0, & \hat{B} \leq t_B. \end{cases}$$

When $\hat{A} > t_A, \hat{B} > t_B$, and $\hat{d} > t_{\text{det}}$, the unrestricted empirical joint rule is

$$\hat{\lambda}_{\text{joint}} = \frac{\hat{E}\hat{D} - \hat{B}\hat{C}}{\hat{d}}, \quad \hat{\omega}_{\text{joint}} = \frac{\hat{A}\hat{D} - \hat{E}\hat{C}}{\hat{d}}.$$

For the prespecified nonempty compact box $\mathcal{B}_\beta = [L_\lambda, U_\lambda] \times [L_\omega, U_\omega]$ containing the origin, the boundary-correct empirical choice is

$$\hat{\beta}_{\mathcal{B}_\beta} \in \arg \min_{\beta \in \mathcal{B}_\beta} \left\{ 2\hat{\ell}^\top \beta + \beta^\top \hat{\mathbf{H}}_{\text{ATE}} \beta \right\}.$$

If the unrestricted joint rule lies outside $\mathcal{B}_\beta$, this quadratic program must be solved rather than clipping its coordinates separately; rectangular coordinate-wise clipping is exact only after imposing $\hat{E} = 0$. For the joint rule, if a required threshold fails or the quadratic solver is numerically unstable, the prespecified fallback is $(0, 0)$; the axis rule uses its coordinate-specific threshold definitions above.

The population-orthogonality plug-in imposes $\hat{E} = 0$ and uses $(-\hat{C}/\hat{A}, \hat{D}/\hat{B})$. The empirical joint minimizer retains $\hat{E}$ and applies Equation (21). The two rules differ because sample orthogonality is not exact.

Each sample covariance is unbiased for its corresponding fixed-score population moment, but the ratios defining $\hat{\lambda}$ and $\hat{\omega}$ are nonlinear and are not generally unbiased for the population oracle coefficients. The thresholds and feasible box control instability; they do not turn these selected coefficients into finite-sample oracle estimates.

The displayed analytic ratios apply only when their corresponding denominator thresholds pass. Imposing $\hat{E} = 0$ does not assert that the realized tuning covariance vanishes. It applies the proven population restriction $E = 0$ before solving the sample problem. Retaining $\hat{E} = s_R(\Delta, g)$ instead minimizes the realized empirical quadratic, whose interaction term is generally nonzero because of tuning-sample noise.

Algorithm 1 states the exact three-way implementation.

Algorithm 1: Three-way honest ATE estimation

Input: Independent RCT and OBS samples; a prespecified feasible box and denominator thresholds.

- 1 **Split:** Divide each source into nuisance-training, coefficient-tuning, and final-evaluation blocks.
- 2 **Fit:** Estimate $\hat{\mu}_R$, $\hat{\mu}_O$, g , and, if required, a transport model without using any tuning or evaluation observations. The exact theorem uses the true ratio r ; replacing it by $\hat{r}$ defines the approximate regime described below.
- 3 **Tune:** On the independent tuning blocks, compute the five moments, apply the prespecified denominator thresholds, and solve the box-constrained quadratic. Use $(0, 0)$ as the declared fallback when the moments are unstable.
- 4 **Evaluate:** Hold every fitted object fixed and evaluate Equation (12) on the untouched final blocks.
- 5 **Infer:** Set $\psi_R = Z_{\hat{\lambda}} - \hat{\omega}g$ and $\psi_O = \hat{\omega}rg$, and report $\hat{V} = s_R^2(\psi_R)/n + s_O^2(\psi_O)/N$. Under conditional Lindeberg conditions, bounded conditional $(2 + \delta)$ moments, and a nondegenerate variance limit, use $\hat{\theta}_r \pm z_{1-\alpha/2} \sqrt{\hat{V}}$.

Output: $\hat{\theta}_r$, $\hat{V}$, and the corresponding Wald interval.

For explicit evaluation notation, define

$$\psi_{R,i} = Z_{\hat{\lambda}}(V_i^R) - \hat{\omega}g(X_i^R), \quad \psi_{O,j} = \hat{\omega}r(X_j^O)g(X_j^O),$$

and let $s_{R,\text{ev}}^2$ and $s_{O,\text{ev}}^2$ be the usual sample variances with denominators $n-1$ and $N-1$, respectively. For $n, N \geq 2$, the fixed-selected-coefficient variance estimator and Wald interval are

$$\hat{V} = \frac{s_{R,\text{ev}}^2(\psi_R)}{n} + \frac{s_{O,\text{ev}}^2(\psi_O)}{N},$$

$$\hat{\theta}_r \pm z_{1-\alpha/2} \sqrt{\hat{V}}.$$

Conditional unbiasedness of $\hat{V}$ follows from evaluation-block independence. Wald coverage additionally requires the stated conditional Lindeberg, moment, and nondegeneracy conditions; it is not a finite-sample coverage statement. Here $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of the standard normal distribution; for $\alpha = 0.05$, it is approximately 1.96.

Proposition 2 (Exact variance estimation after honest selection). *Under Theorem 1, let $(\hat{\lambda}, \hat{\omega})$ be $\mathcal{F}$ -measurable and suppose $n, N \geq 2$. Conditional on $\mathcal{F}$, the selected estimator is unbiased and*

$$\mathbb{E}(\hat{V} \mid \mathcal{F}) = \text{Var}\{\hat{\theta}_r(\hat{\lambda}, \hat{\omega}) \mid \mathcal{F}\}.$$

Theorem 3 (Conditional central limit theorem and Wald inference). *In addition to Proposition 2, suppose $n, N \rightarrow \infty$, $n/N \rightarrow \kappa \in [0, \infty)$, and for some $\delta > 0$ the centered RCT and OBS final-score arrays have uniformly bounded conditional $(2 + \delta)$ moments. This moment*

assumption supplies the required conditional Lindeberg condition. Assume also that

$$n \text{Var}\{\hat{\theta}_r(\hat{\lambda}, \hat{\omega}) \mid \mathcal{F}\} \rightarrow \sigma^2 \in (0, \infty)$$

in probability. Then

$$\frac{\hat{\theta}_r - \theta}{\sqrt{\hat{V}}} \rightsquigarrow N(0, 1),$$

and the displayed Wald interval has asymptotic coverage $1 - \alpha$. Variance unbiasedness alone does not imply this coverage statement.

With an estimated ratio $\hat{r}$, define

$$h_n(X) = \{\hat{r}(X) - r(X)\}g(X). \quad (27)$$

The exact conditional drift is

$$\omega b_{\mathcal{F}} = \omega \mathbb{E}_O\{h_n(X)\}. \quad (28)$$

Honest splitting prevents evaluation leakage but does not remove this drift. A direct sufficient negligible-error regime is

$$\sqrt{n} \omega \mathbb{E}_O\{h_n(X)\} = o_p(1), \quad |\omega| \sqrt{\frac{n}{N}} \|h_n - \mathbb{E}_O(h_n)\|_{L^2(P_O)} = o_p(1). \quad (29)$$

Evaluation-only variance is first-order adequate under these conditions and the preceding CLT assumptions. Otherwise the ratio learner's influence-function contribution, the OBS empirical-process term, and their covariances must be included.

Estimating the density ratio. All three constructions below use only a ratio-training block and are held fixed before coefficient tuning and final evaluation. Let $\mathbb{P}_{R,\text{rat}}$ and $\mathbb{P}_{O,\text{rat}}$ denote empirical averages on its RCT and OBS parts. First, pool the ratio-training covariates and let $S = 1$ indicate an RCT observation and $S = 0$ an OBS observation. With $\pi = P(S = 1)$ and $s(x) = P(S = 1 \mid X = x)$, Bayes' rule gives

$$r(x) = \frac{1 - \pi}{\pi} \frac{s(x)}{1 - s(x)}, \quad \hat{r}_{\text{cls}}(x) = \frac{1 - \hat{\pi}}{\hat{\pi}} \frac{\hat{s}(x)}{1 - \hat{s}(x)}.$$

For an unweighted pooled classifier with ratio-training sizes m_R^{rat} and m_O^{rat} , $\hat{\pi} = m_R^{\text{rat}} / (m_R^{\text{rat}} + m_O^{\text{rat}})$. Class weighting or resampling requires replacing this factor by the effective training prior. The odds representation also requires overlap, $0 < s(X) < 1$, on the RCT covariate support. Probability clipping stabilizes the estimate but changes it from the exact density ratio.

A Riesz construction estimates the transport functional directly (Chernozhukov et al., 2021, 2022). Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$ be a fixed feature map and define

$$\widehat{M}_O = \mathbb{P}_{O,\text{rat}}\{\phi(X)\phi(X)^\top\}, \quad \widehat{b}_R = \mathbb{P}_{R,\text{rat}}\{\phi(X)\}.$$

For $\rho > 0$, the empirical Riesz quadratic and its closed-form solution are

$$\begin{aligned}\hat{\eta}_\rho &= \arg \min_{\eta \in \mathbb{R}^p} \left[\eta^\top \widehat{M}_O \eta - 2\widehat{b}_R^\top \eta + \rho \|\eta\|_2^2 \right] = (\widehat{M}_O + \rho I_p)^{-1} \widehat{b}_R, \\ \widehat{r}_{\text{Riesz}}(x) &= \phi(x)^\top \widehat{\eta}_\rho.\end{aligned}$$

At $\rho = 0$, a compatible singular system uses $\widehat{M}_O^+ \widehat{b}_R$. A restricted feature map estimates a projected representer, and the resulting weights need not be nonnegative.

A nonnegative automatic-balancing alternative starts with an out-of-sample-evaluable family $w_\eta : \mathcal{X} \rightarrow [0, \infty)$. For a fixed feature map ϕ , a positive semidefinite matrix W , and $\gamma > 0$, define $\|v\|_W^2 = v^\top W v$ and solve

$$\begin{aligned}\widehat{\eta}_{\text{bal}} &\in \arg \min_{\eta: \mathbb{P}_{O,\text{rat}} w_\eta = 1} \left[\|\mathbb{P}_{O,\text{rat}}\{w_\eta(X)\phi(X)\} - \mathbb{P}_{R,\text{rat}}\{\phi(X)\}\|_W^2 + \gamma \mathbb{P}_{O,\text{rat}}\{w_\eta(X)^2\} \right], \\ \widehat{r}_{\text{bal}}(x) &= w_{\widehat{\eta}_{\text{bal}}}(x).\end{aligned}$$

Including g among the coordinates of ϕ directly targets the transported moment used by the ATE correction. The dispersion penalty controls unstable weights but generally prevents exact empirical balance.

These constructions do not automatically inherit the exact-r theorem. For any fixed evaluation-independent weight function w , replacing r by w gives

$$\begin{aligned}\mathbb{E}\{\widehat{\theta}_w(\lambda, \omega) \mid \mathcal{F}\} &= \theta + \omega\{P_O(wg) - P_R(g)\} \\ &= \theta + \omega P_O\{(w - r)g\}.\end{aligned}$$

Thus exact conditional unbiasedness holds for $w = r$, or more generally for a fixed w satisfying the exact population balance condition $P_O(wg) = P_R(g)$. Classifier, restricted Riesz, and automatic-balancing estimates are otherwise in the estimated-ratio regime governed by the drift and rate conditions above. The Auto-DML citation motivates the Riesz learning criterion; it does not by itself make the present score Neyman-orthogonal to ratio error or supply the ratio learner's first-order variance contribution.

Nested outer cross-fitting. To reuse observations without scoring any observation with a nuisance fit trained on its own outcome, prespecify paired outer folds in both sources, or randomize the folds independently and include that randomness in the conditioning field. Within each outer complement, use inner splits to fit nuisances and select $(\widehat{\lambda}_k, \widehat{\omega}_k)$; then score only the untouched paired folds. With fixed nonnegative weights a_k summing to one, define

$$\widehat{\theta}_k = \mathbb{P}_{R,k} Z_{\widehat{\lambda}_k}^{(-k)} + \widehat{\omega}_k \left\{ \mathbb{P}_{O,k}(r g^{(-k)}) - \mathbb{P}_{R,k} g^{(-k)} \right\}, \quad \widehat{\theta}_{\text{outer}} = \sum_k a_k \widehat{\theta}_k.$$

Proposition 3 (What outer-fold honesty proves). *Under the common covariate marginal or exact r , each $\widehat{\theta}_k$ is conditionally unbiased given its own outer complement, and $\widehat{\theta}_{\text{outer}}$ is unconditionally unbiased. Overlapping outer training complements make the fold estimators dependent, so these facts do not justify adding foldwise variances or using a naive pooled*

Wald interval.

A pooled cross-fitted interval requires an additional stability or influence-function argument, for example convergence of fold-specific RCT and OBS scores to common deterministic limits in their respective L_2 norms, uniform $(2 + \delta)$ moments, and a nondegenerate two-sample limit. The present theory does not assert a general cross-fitted variance theorem without such conditions.

Diagnostics. An implementation should report split assignments, propensity bounds, candidate boxes, all denominator and determinant thresholds, coefficient fallbacks, ratio overlap and clipping, and the selected RCT-only comparator. Near-zero $\hat{A}$, $\hat{B}$, or $\hat{d}$ must trigger the prespecified rule rather than an unstable analytic ratio.

4. Prediction-Powered Fusion Learning for CATE

4.1. Prediction-powered loss, coefficient roles, and the ATE relation

For a fixed candidate CATE function t , define the density-ratio-weighted prediction-powered squared loss

$$\hat{L}_{\text{DR}}(t; \lambda, \omega) = \mathbb{P}_{R,n}\{Z_\lambda - t(X)\}^2 \quad (30)$$

$$+ \omega \left[\mathbb{P}_{O,N}\{r(X)(g(X) - t(X))^2\} - \mathbb{P}_{R,n}\{(g(X) - t(X))^2\} \right]. \quad (31)$$

For every fixed candidate considered below, assume $Z_\lambda, t \in L^2(P_R)$, and let $G_t(X) = \{g(X) - t(X)\}^2$ satisfy $G_t \in L^1(P_R)$ and $rG_t \in L^1(P_O)$.

Proposition 4 (Exact target of the prediction-powered CATE loss). *Under Assumptions 1, 3, and 4,*

$$\mathbb{E}\{\hat{L}_{\text{DR}}(t; \lambda, \omega)\} = Q(\lambda) + \mathbb{E}_R\{t(X) - \tau(X)\}^2, \quad (32)$$

$$Q(\lambda) = \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2. \quad (33)$$

The common-marginal case sets $r = 1$. In both cases, the expected loss is flat in ω .

Because $Z_\lambda = Z_0 + \lambda\Delta$ and $\mathbb{E}_R(\Delta | X) = 0$,

$$Q(\lambda) = Q(0) + 2C\lambda + A\lambda^2.$$

When $A > 0$, define the integrated pseudo-outcome-noise oracle

$$\lambda_Q^* = \arg \min_{\lambda \in \mathbb{R}} Q(\lambda) = -C/A. \quad (34)$$

Write $\lambda_{\text{ATE}}^* := \lambda^*$ and $\omega_{\text{ATE}}^* := \omega^*$ for the two coordinates of the population-orthogonal ATE variance oracle in Equation (24). Then $\lambda_Q^* = \lambda_{\text{ATE}}^*$. When $A = 0$, $\Delta = 0$ almost surely and $Q(\lambda)$ is flat in λ . The ATE variance coefficient $\omega_{\text{ATE}}^* = D/B$ is a possible training default, but Equation (33) does not identify it as a CATE-risk optimizer.

Proposition 5 (Fixed measurable and pointwise noise coefficients). *Under Assumptions 1 and 4, let $\lambda(\cdot)$ be a fixed measurable function such that $Z_0 + \lambda(X)\Delta$ is integrable. Then*

$$\mathbb{E}_R\{Z_0 + \lambda(X)\Delta \mid X\} = \tau(X).$$

For the pointwise statement, choose measurable versions of the displayed conditional covariance and variance and assume they are finite. For P_R^X -almost every x on which $\text{Var}_R(\Delta \mid X = x) > 0$, the conditional-noise minimizer is

$$\lambda_Q^*(x) = -\frac{\text{Cov}_R(Z_0, \Delta \mid X = x)}{\text{Var}_R(\Delta \mid X = x)}.$$

On the zero-denominator set the conditional criterion is flat, and any finite measurable value may be assigned. The ratio must be measurable on the positive-denominator set, and the resulting pseudo-outcome must remain integrable.

The global scalar λ_Q^* , the pointwise function $\lambda_Q^*(x)$, and the coefficient chosen by a restricted or penalized learner are different oracle objects.

The sampling variance of a fixed candidate loss identifies a distinct coefficient. Let $U_{t,\lambda} = \{Z_\lambda - t(X)\}^2$.

Proposition 6 (Fixed-loss variance oracle). *Under Assumption 4, fix $\mathcal{F}$ -measurable t and λ ; the established conditioning convention also makes g and r $\mathcal{F}$ -measurable. Suppose $U_{t,\lambda} \in L^2(P_R)$, $G_t \in L^2(P_R)$, and $rG_t \in L^2(P_O)$. If the denominator below is positive, the unrestricted variance-minimizing coefficient is*

$$\omega_{\text{LV}}^*(t; \lambda) = \frac{\text{Cov}_R(U_{t,\lambda}, G_t)}{\text{Var}_R(G_t) + (n/N) \text{Var}_O(rG_t)}. \quad (35)$$

Under the common covariate marginal this denominator is $(1 + n/N) \text{Var}_R(G_t)$. If the denominator is zero, the empirical correction is constant and its variance criterion is flat in ω .

This oracle minimizes the variance of one fixed empirical loss. It does not minimize the final learned CATE risk. The square-integrability conditions are fourth-moment-type requirements on the underlying residuals.

Final learner-risk oracle. Let $\mathfrak{C}$ be a prespecified finite candidate class. An index

$$c = (\lambda_c, \omega_c, p_c, \rho_c) \in \mathfrak{C} \quad (36)$$

specifies the two fusion coefficients, the sieve dimension, and the penalty level. All other components of the learning rule, including the data splits, nuisance algorithms, basis construction, and optimization convention, are also prespecified. Let $\mathcal{D}_{\text{tr}}$ collect all pre-validation randomness, including the candidate-training samples, nuisance and ratio training samples, basis fits, and independently randomized split assignments, and set $\mathcal{F}_{\text{train}} = \sigma(\mathcal{D}_{\text{tr}})$. Let $\hat{t}_c^{\mathcal{D}_{\text{tr}}}$ denote the resulting

fitted function and define its sample-specific target risk and its algorithm-induced expected risk by

$$\mathcal{R}_{\mathcal{D}_{\text{tr}}}(c) = \mathbb{E}_R \left[\{ \hat{t}_c^{\mathcal{D}_{\text{tr}}}(X) - \tau(X) \}^2 \middle| \mathcal{D}_{\text{tr}} \right], \quad (37)$$

$$\mathfrak{R}(c) = \mathbb{E}_{\mathcal{D}_{\text{tr}}} \{ \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c) \}. \quad (38)$$

When the minimum is attained, define the sample-specific final learner-risk oracle

$$c_{\text{risk}}^*(\mathcal{D}_{\text{tr}}) \in \arg \min_{c \in \mathfrak{C}} \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c).$$

This oracle chooses among fitted learning algorithms. It is distinct from both λ_Q^* and $\omega_{\text{LV}}^*(t; \lambda)$.

To make the fallback comparison exact, let

$$\mathfrak{C}_R = \{(0, 0, p, \rho) : (p, \rho) \in \mathcal{H}_R\} \subseteq \mathfrak{C} \quad (39)$$

contain the identical RCT-only learners over the prespecified sieve and penalty grid $\mathcal{H}_R$. Identical means that the RCT observations, folds, RCT nuisance fits, basis, penalty scaling, and optimization code agree with the corresponding fusion candidates; only the two OBS fusion channels are set to zero.

Proposition 7 (Candidate-class oracle dominance). *For every realized candidate-training sample for which the displayed infima are finite,*

$$\inf_{c \in \mathfrak{C}} \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c) \leq \inf_{c \in \mathfrak{C}_R} \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c). \quad (40)$$

Likewise,

$$\inf_{c \in \mathfrak{C}} \mathfrak{R}(c) \leq \inf_{c \in \mathfrak{C}_R} \mathfrak{R}(c). \quad (41)$$

If the infima are attained, they define the sample-specific and expected-training final CATE-risk oracles. Either inequality is strict only when at least one fusion candidate has risk strictly below the optimally tuned RCT-only class under the corresponding comparison.

Proposition 7 is an infeasible oracle statement because its risks use the unknown τ . It does not say that a data-adaptive selector dominates the RCT-only learner on every finite sample.

Common-reference honest validation. Condition on the pre-validation sigma-field $\mathcal{F}_{\text{train}}$ defined above. Construct one shared RCT-valid pseudo-outcome $Z_{\lambda_{\text{ref}}}$ on an untouched RCT validation block using a fixed reference coefficient. For candidate c , set

$$G_c(X) = \{g(X) - \hat{t}_c^{\mathcal{D}_{\text{tr}}}(X)\}^2. \quad (42)$$

Let $r^\dagger = 1$ under the common covariate marginal, $r^\dagger = r$ under exact covariate transport, and $r^\dagger = \hat{r}$ in the estimated-ratio regime. Define the shared-reference validation score

$$\begin{aligned} \hat{\mathcal{R}}_c^{\text{ref}} &= \mathbb{P}_{R, \text{val}} \left[\{Z_{\lambda_{\text{ref}}} - \hat{t}_c^{\mathcal{D}_{\text{tr}}}(X)\}^2 \right] \\ &\quad + \mathbb{P}_{O, \text{val}} \{r^\dagger(X) G_c(X)\} - \mathbb{P}_{R, \text{val}} G_c(X). \end{aligned} \quad (43)$$

All candidates must use the same reference pseudo-outcome, g , ratio rule, and validation observations. Under the common marginal or exact r , known RCT propensity, and honest conditioning,

$$\mathbb{E}\{\widehat{\mathcal{R}}_c^{\text{ref}} \mid \mathcal{F}_{\text{train}}\} = Q(\lambda_{\text{ref}}) + \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c). \quad (44)$$

Thus candidate comparisons remove the same additive reference noise. The OBS correction is not required for validity of this identity, but it can reduce the sampling variability of validation when the prediction is useful.

For the approximate regime, define the conditional reference bias

$$b_{\text{ref}}(X) = \mathbb{E}_R\{Z_{\lambda_{\text{ref}}} - \tau(X) \mid X, \mathcal{F}_{\text{train}}\}, \quad (45)$$

the ratio and nuisance drifts

$$\delta_{r,c} = \mathbb{E}_O[\{r^\dagger(X) - r(X)\}G_c(X)], \quad (46)$$

$$\delta_{\text{nuis},c} = 2\mathbb{E}_R[b_{\text{ref}}(X)\{\tau(X) - \widehat{t}_c^{\mathcal{D}_{\text{tr}}}(X)\}], \quad (47)$$

and their uniform magnitudes

$$\Delta_r = \sup_{c \in \mathfrak{C}} |\delta_{r,c}|, \quad \Delta_{\text{nuis}} = \sup_{c \in \mathfrak{C}} |\delta_{\text{nuis},c}|. \quad (48)$$

Known trial propensity and an honest reference nuisance give $b_{\text{ref}} = 0$ for every outcome-regression fit. Exact transport gives $\Delta_r = 0$. Estimated ratios or estimated propensities must instead be controlled through the displayed drifts or a stronger problem-specific expansion. Adaptive reuse of validation observations violates the held-out product law and measurability used here; it is outside Proposition 8 unless a separate selective-inference or stability theorem is supplied.

Proposition 8 (Validation-selector oracle inequality). *Let*

$$\widehat{c} \in \arg \min_{c \in \mathfrak{C}} \widehat{\mathcal{R}}_c^{\text{ref}} \quad (49)$$

and define the conditional validation fluctuation

$$\varepsilon_{\text{val}} = \sup_{c \in \mathfrak{C}} \left| \widehat{\mathcal{R}}_c^{\text{ref}} - \mathbb{E}\{\widehat{\mathcal{R}}_c^{\text{ref}} \mid \mathcal{F}_{\text{train}}\} \right|. \quad (50)$$

Then

$$\mathcal{R}_{\mathcal{D}_{\text{tr}}}(\widehat{c}) \leq \inf_{c \in \mathfrak{C}} \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c) + 2\varepsilon_{\text{val}} + 2\Delta_r + 2\Delta_{\text{nuis}} \quad (51)$$

and therefore the same upper bound holds with the infimum over $\mathfrak{C}$ replaced by the larger infimum over $\mathfrak{C}_R$.

If $\varepsilon_{\text{val}} + \Delta_r + \Delta_{\text{nuis}} = o_p(1)$, Proposition 8 gives asymptotic weak noninferiority relative to the identical optimally tuned RCT-only class. Passing from this conditional statement to expected-training risk additionally requires uniform integrability. Strict asymptotic improvement requires a positive oracle gap that dominates the validation and drift remainders. No part of this argument gives universal finite-sample or realized-sample dominance.

Why λ_Q^* need not optimize sieve risk. Consider an unpenalized linear sieve fitted on RCT pseudo-outcomes with $\omega = 0$, basis b_p , population Gram matrix $G_p = \mathbb{E}_R\{b_p(X)b_p(X)^\top\}$, projection t_p of τ , and leverage $h_p(X) = b_p(X)^\top G_p^{-1} b_p(X)$. Under the usual fixed-sieve least-squares expansion, the λ -dependent part of the leading integrated estimation variance is

$$\frac{1}{n} \mathbb{E}_R [h_p(X) \text{Var}_R(Z_\lambda | X)]. \quad (52)$$

The oracle λ_Q^* minimizes the same expression without $h_p(X)$. The two coefficients coincide when the leverage is constant, or under another condition that makes the two weighted quadratics share a minimizer, but they need not coincide in general. Penalties, ω , and data-adaptive basis selection create further differences between the integrated-noise oracle and the final learner-risk oracle.

For $\omega \neq 1$, define $\tilde{Z}_{\lambda,\omega} = (Z_\lambda - \omega g)/(1 - \omega)$.

Proposition 9 (Exact square completion). *For every $\omega \neq 1$,*

$$\{Z_\lambda - t(X)\}^2 - \omega\{g(X) - t(X)\}^2 = (1 - \omega)\{t(X) - \tilde{Z}_{\lambda,\omega}\}^2 - \frac{\omega}{1 - \omega}\{Z_\lambda - g(X)\}^2.$$

Consequently,

$$\hat{L}_{\text{DR}}(t; \lambda, \omega) = (1 - \omega) \mathbb{P}_{R,n} \{t(X) - \tilde{Z}_{\lambda,\omega}\}^2 \quad (53)$$

$$+ \omega \mathbb{P}_{O,N} \{r(X)(g(X) - t(X))^2\} + \left[-\frac{\omega}{1 - \omega} \mathbb{P}_{R,n} \{Z_\lambda - g(X)\}^2 \right]. \quad (54)$$

The bracketed term is constant in t .

For $0 \leq \omega < 1$, the two t -dependent terms form a nonnegative mixture of squared losses. At $\omega = 1$, the transformed outcome is undefined and the original objective must be used. For $0 \leq \omega \leq 1$, the original empirical quadratic is positive semidefinite in unrestricted fitted values and in the coefficients of a linear sieve because its Hessian is a nonnegative weighted sum of Gram matrices. Outside this interval, or for nonlinear parameterizations, convexity is not guaranteed. Under a penalty or parameter constraint, rescaling the completed loss changes the relative penalty scale, so the original quadratic and its stated penalty must define the estimator.

4.2. Implementable F-Learner and prospective R-loss extension

We use *F-Learner* as a provisional working label only for CATE estimators that minimize Equation (31), which is built from the RCT-valid AIPW pseudo-outcome Z_λ . The label does not claim name novelty and does not denote a generic unification of doubly robust and R losses.

For a linear sieve $t_\beta(x) = b(x)^\top \beta$, with $b(x) \in \mathbb{R}^p$, add the penalty $\rho \beta^\top P \beta$, where $\rho \geq 0$ and $P = P^\top \succeq 0$ belongs to $\mathbb{R}^{p \times p}$. Define

$$G_R = \mathbb{P}_{R,n} \{b(X)b(X)^\top\}, \quad G_{O,r} = \mathbb{P}_{O,N} \{r(X)b(X)b(X)^\top\}, \quad (55)$$

$$h_\lambda = \mathbb{P}_{R,n} \{b(X)Z_\lambda\}, \quad d_g = \mathbb{P}_{O,N} \{r(X)b(X)g(X)\} - \mathbb{P}_{R,n} \{b(X)g(X)\}. \quad (56)$$

Whenever the matrix is invertible, the penalized F-Learner solution is

$$\hat{\beta}_{\lambda,\omega} = \{(1 - \omega)G_R + \omega G_{O,r} + \rho P\}^{-1} \{h_\lambda + \omega d_g\}. \quad (57)$$

Because h_λ is affine in λ , a fixed- ω candidate path can be computed efficiently. A frozen neural basis replaces $b(x)$ by a representation learned without the candidate-training or validation observations and inherits Equation (57) conditional on that representation. End-to-end neural fitting is nonconvex, and the present linear-sieve guarantees do not cover it.

Out-of-fold nuisance and candidate construction. Partition the RCT and OBS indices into prespecified folds. For each held-out fold k , fit $\hat{\mu}_R^{(-k)}$ and $\hat{\mu}_O^{(-k)}$ without that fold, use the same

$$g^{(-k)}(x) = \hat{\mu}_{O,1}^{(-k)}(x) - \hat{\mu}_{O,0}^{(-k)}(x)$$

on held-out RCT and OBS covariates, and construct $Z_{0,i}^{(-k)}$, $Z_{1,i}^{(-k)}$, and $Z_{\lambda,i}^{(-k)} = Z_{0,i}^{(-k)} + \lambda\{Z_{1,i}^{(-k)} - Z_{0,i}^{(-k)}\}$ only on the held-out RCT fold. This prevents an observation's own outcome from entering its nuisance prediction, but estimates from different folds remain dependent.

For each $c \in \mathfrak{C}$, candidate fitting, basis construction, penalty selection, and tuning-dependent preprocessing must exclude the shared validation observations. The common-reference validation score then ranks already-fitted candidates. Reusing validation outcomes to fit a candidate or its preprocessing violates the held-out product law and lies outside Proposition 8.

A prospective R-loss extension can be implemented as an empirical candidate, but it requires a separate theorem before it belongs to the F-Learner class. Let m be a generic fixed, cross-fitted marginal-outcome regression trained independently of the observations on which the following loss is evaluated. One candidate objective is

$$\hat{L}_R(t; \omega) = \mathbb{P}_{R,n} [\{Y - m(X)\} - \{A - e(X)\}t(X)]^2 \quad (58)$$

$$+ \omega \left[\mathbb{P}_{O,N} \{r(X)(g(X) - t(X))^2\} - \mathbb{P}_{R,n} \{(g(X) - t(X))^2\} \right], \quad (59)$$

Randomized treatment implies

$$\mathbb{E}_R[(A - e(X))\{Y - m(X)\} \mid X] = e(X)\{1 - e(X)\}\tau(X) \quad (60)$$

for every fixed m . This identity motivates the residualized objective. No λ channel has yet been derived for this R-loss candidate. A complete result must still derive the precise target under penalties and restricted learners, a principled role for λ , an honest validation score, and estimated-nuisance rates. We therefore present Equation (59) as a planned extension rather than a proved contribution.

Under covariate shift, an exact ratio can be recovered from domain-classification odds. If $S = 1$ denotes RCT membership, $\pi = P(S = 1)$, and $s(x) = P(S = 1 \mid X = x)$, then

$$r(x) = \frac{1 - \pi}{\pi} \frac{s(x)}{1 - s(x)}. \quad (61)$$

An estimated ratio $\hat{r}$ replaces exact target preservation with an explicit drift. Clipping changes the exact transported target and must be reported.

A loss-targeted alternative chooses a nonnegative, out-of-sample-evaluable weight function w to balance

$$\mathcal{G}(\mathcal{T}; g) = \{(g - t)^2 : t \in \mathcal{T}\}. \quad (62)$$

If $\mathbb{E}_O\{w(g - t)^2\} = \mathbb{E}_R\{(g - t)^2\}$ for every $t \in \mathcal{T}$, Equation (33) remains exact with w in place of r . For a linear sieve, balancing bg and $bb^\top$ preserves the minimizer; balancing g^2 is additionally required only to preserve the absolute loss value. Exact balancing on the same observations used to fit t makes the correction zero or constant in the sieve coefficient. Weight learning and candidate fitting must therefore use different honest blocks or appropriate outer folds.

For approximate weights, define

$$\Delta_{\mathcal{T}}(w) = \sup_{t \in \mathcal{T}} \left| \mathbb{E}_O\{w(X)(g(X) - t(X))^2\} - \mathbb{E}_R\{(g(X) - t(X))^2\} \right|. \quad (63)$$

The uniform population-loss drift is bounded by $|\omega|\Delta_{\mathcal{T}}(w)$, and the excess target risk of the weighted population minimizer is at most $2|\omega|\Delta_{\mathcal{T}}(w)$ when both minimizers exist.

Current scope and open theory. The proved CATE statements cover the AIPW-pseudo-outcome squared loss with fixed honest nuisances and either a common covariate marginal or exact transport. The linear-sieve solution and frozen-feature extension are conditional results. Estimated-ratio rates, adaptive loss-targeted balancing, end-to-end neural optimization, the R-loss extension, and cross-fitted inference remain open theory. Honest splitting prevents direct reuse but does not by itself close these gaps.

A reproducible analysis should report the candidate class $\mathfrak{C}$, RCT-only subclass $\mathfrak{C}_R$, folds, basis dimension, penalty and scaling, ratio method and overlap diagnostics, denominator fallbacks, selected candidate, and validation score of the identical RCT-only comparator. Values of ω outside $[0, 1]$ require an explicit curvature check of the original empirical objective.

5. Experiments

5.1. Completed ATE benchmarks

The completed experiments evaluate ATE estimation only. All nested benchmarks use 20 Monte Carlo repetitions per cell, an RCT-only AIPW baseline, a λ -only outcome-model fusion estimator, an ω -only PPI correction estimator, and the joint estimator. The code retains two stale internal names: `haipw_only` denotes the λ -only estimator and `ppi_power_tuned_only` denotes the ω -only estimator. The first label does not identify the method with the separate HAIPW literature.

The fixed-OBS-budget benchmark contains 16 design cells: three arbitrary nonlinear SCMs and one STAR real- X construction, each under a common covariate marginal and covariate shift and $n_R \in \{20, 30\}$, with $N_O = 5000$. The joint estimator has the smallest RMSE in 11 cells, and the λ -only estimator is best in the remaining five. The joint estimator has lower RMSE than AIPW in all 16 cells, with mean RMSE ratio 0.648 and mean empirical-variance ratio 0.439 relative to AIPW. The λ -only estimator also beats AIPW in all 16 cells, with mean RMSE ratio 0.679. The ω -only estimator is not best in any cell and has mean RMSE ratio 1.005.

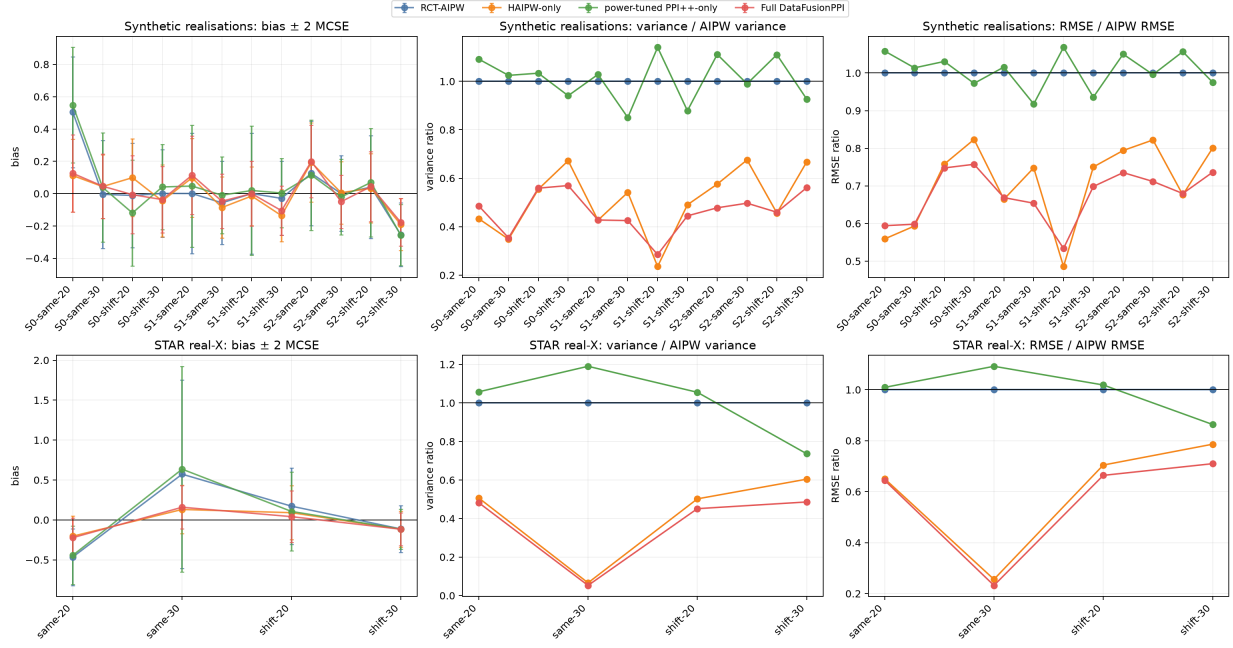


Figure 1: Fixed-OBS-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30\}$ and $N_O = 5000$.

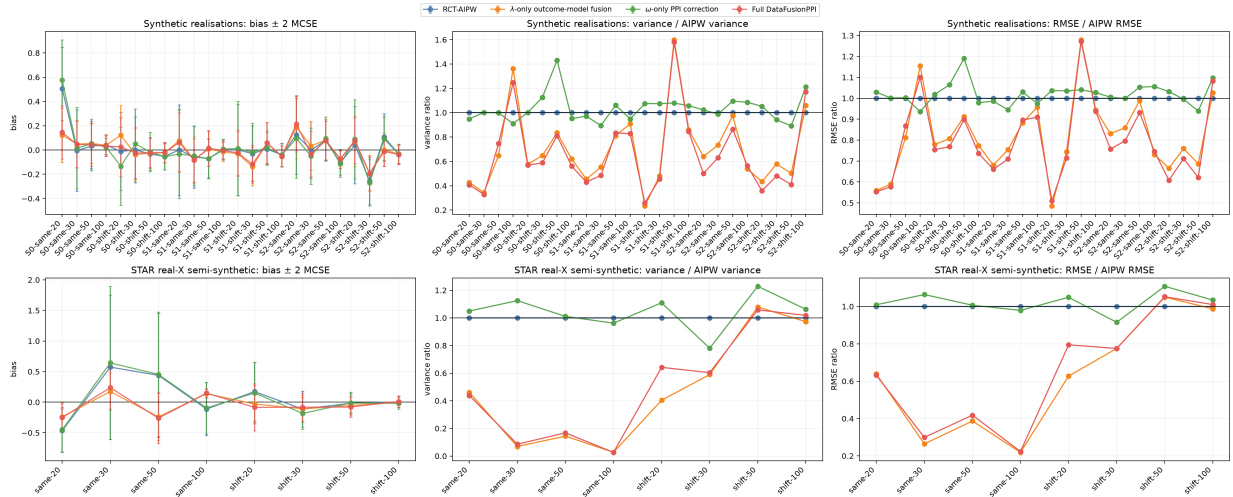


Figure 2: Proportional-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$.

The proportional-budget benchmark contains 32 cells with $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$. The joint estimator has the smallest RMSE in 18 cells, the λ -only estimator in 10, the ω -only estimator in one, and AIPW in three. The joint estimator has lower RMSE than AIPW in 27 of 32 cells, with mean RMSE ratio 0.760 and mean empirical-variance ratio 0.625. The λ -only estimator beats AIPW in 28 cells and has mean RMSE ratio 0.769. The ω -only estimator beats AIPW in 10 cells and has mean RMSE ratio 1.020. These counts show that joint tuning is often useful but is not uniformly best after coefficient estimation.

The STAR benchmark uses only real kindergarten covariates from Project STAR. Treatment assignments, latent confounding, potential outcomes, and effect heterogeneity are generated synthetically. It is therefore a real- X , semi-synthetic benchmark rather than an analysis of STAR’s original intervention or outcomes. The planned WHI CT+OS analysis has not been run because participant-level access, a semantic crosswalk, and a survival or censoring extension are still required.

5.2. Planned CATE evaluation and interpretation

CATE experiments are planned. The primary design will compare the proved AIPW-pseudo-outcome F-Learner with the prospective R-loss extension using additive cubic B-splines, frozen neural bases, and linear sieves under a common covariate marginal and covariate shift. Each study will compare trial-only, λ -only, ω -only, and joint tuning using held-out RCT-population CATE risk. Classifier-only, balancing-only, hybrid, and oracle transport weights will be separated. No CATE result is claimed from Figures 1 and 2.

The empirical conclusion is deliberately limited. The two-channel population oracle weakly improves the ATE variance objective when the trial-only coefficient is feasible, but estimated coefficients do not inherit a universal no-harm property. The present 20-repetition studies suggest that the λ channel is the more stable source of gain and that joint tuning can add value in some designs. More Monte Carlo repetitions, completed CATE experiments, additional real- X supports, and an authorized real-outcome analysis are required before making a stronger empirical claim.

5.3. Scope, failure modes, and conclusion

The framework uses observational information through prediction rather than by declaring the observational treatment contrast causal. Trial randomization validates the pseudo-outcome for every fixed outcome regression. A common covariate marginal or an exact density ratio centers the ATE control variate and the prediction-powered CATE loss, which creates separate λ - and ω -channels while retaining RCT-population targets.

The first boundary is covariate transport. Without weighting under $P_R^X \neq P_O^X$, the ATE correction has drift $\omega\{P_O g - P_R g\}$, and the unweighted CATE loss targets the wrong covariate average. Exact $r = dP_R^X/dP_O^X$ removes both discrepancies. An estimated ratio instead contributes the explicit ATE drift $\omega P_O\{(\hat{r} - r)g\}$ and the CATE validation drift in Equation (48). Evaluation-only variance is first-order adequate only under the stated negligible-drift conditions; otherwise ratio-training influence and covariance terms are required.

The second boundary is trial propensity overlap. Proposition 1 uses the known, strictly positive randomized-trial propensity. Extreme inverse-propensity factors can make the required moments

unstable even when the conditional-mean identity remains algebraically valid, and an incorrect propensity removes the exact arbitrary- q guarantee.

The third boundary is weak observational prediction. The unrestricted ω -gain depends on how strongly $g(X)$ predicts $\tau(X)$, with attenuation from the finite n/N ratio. The λ -channel gives little gain when $\Delta = Z_1 - Z_0$ is uninformative or nearly degenerate. Population oracles diagnose these limits but do not make their empirical coefficient estimates stable.

The fourth boundary is adaptive reuse. Fitting nuisances, tuning candidates, and evaluating the selected candidate on the same outcomes invalidates the conditioning product law. Three-way splitting prevents direct leakage. Nested outer cross-fitting preserves foldwise and aggregate unbiasedness under exact transport and fold honesty, but overlapping training complements invalidate a naive sum of fold variances. Cross-fitted inference needs an additional stability or influence-function theorem.

The fifth boundary is learner restriction and curvature. Target preservation does not make a finite-sample restricted or highly penalized projection accurate. The square-completed representation is a nonnegative mixture only for $0 \leq \omega < 1$, and positive semidefiniteness at $\omega = 1$ must be read from the original quadratic. Coefficients outside $[0, 1]$, nonlinear parameterizations, and penalty rescaling require direct analysis of the original objective.

The sixth boundary is tail behavior. ATE variance identities require square-integrable scores; the Wald theorem additionally requires conditional Lindeberg control, a sufficient $(2 + \delta)$ -moment condition, and a nondegenerate variance limit. Fixed-loss variance tuning imposes fourth-moment-type requirements. Heavy tails can therefore destabilize oracle formulas even when their algebraic denominators are nonzero.

Within these boundaries, the randomized trial fixes the causal target, the OBS outcome model can reduce within- X pseudo-outcome noise through λ , and the OBS covariate sample can reduce predictable between- X variation through ω . The exact results are conditional identities and population-oracle comparisons. Empirical work must retain the identical trial-only fallback, report selection and denominator instability, and avoid interpreting them as a universal finite-sample no-harm guarantee.

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A. Simulation Details

All completed simulation artifacts were generated with base seed 190602. The primary nested benchmarks use 20 outer Monte Carlo repetitions per design cell and compare four estimators: RCT-only AIPW, λ -only outcome-model fusion, ω -only PPI correction, and joint (λ, ω) fusion. The implementation retains the historical internal codes `haipw_only` and `ppi_power_tuned_only` for the second and third estimators, respectively. These codes are not the user-facing method names, and the first does not refer to the separate HAIPW paper.

The arbitrary simulation family uses three nonlinear structural causal models. In every model, a domain indicator S distinguishes the observational sample ($S = 0$) from the randomized sample ($S = 1$); covariates depend on latent variables, observational treatment depends on both measured covariates and an unmeasured confounder, trial treatment follows the known randomized propensity, and outcomes may depend nonlinearly on covariates, treatment, and the latent confounder. The common-marginal regime uses $P_R^X = P_O^X$. The shifted regime changes the covariate distribution while preserving overlap and estimates the density ratio from a domain classifier. The same estimators and evaluation logic are used in both regimes.

The fixed-OBS-budget experiment has 16 cells: four benchmark families, two covariate regimes, and $n_R \in \{20, 30\}$, with $N_O = 5000$. The proportional experiment has 32 cells: four benchmark families, two regimes, and $n_R \in \{20, 30, 50, 100\}$, with the exact mapping $N_O = 100n_R \in \{2000, 3000, 5000, 10000\}$. Coefficients are estimated by the nested honest procedure; no oracle coefficient or oracle density ratio is used in the reported method comparison. The replication-level CSV files, summaries, and composite plots in `materials/` are the canonical numerical artifacts.

The fourth benchmark family reuses the empirical covariate support from the Tennessee Student/Teacher Achievement Ratio (STAR) experiment. Treatment, latent confounding, outcomes, and treatment-effect heterogeneity are generated by the benchmark code. Consequently this is a real- X , semi-synthetic design, not an evaluation of the original STAR outcomes or intervention effect.

Figure 3 records the earlier two-method pilot. It is retained for provenance but is not the basis for the four-method conclusions in Section 5.

No participant-level WHI CT+OS analysis is included. Such an analysis requires authorized data access, a verified trial-observational semantic crosswalk, a prespecified estimand, and methods appropriate for censoring if the selected outcome is time-to-event. No completed CATE experiment or real-outcome result is represented by the present artifacts.

Illustrative random-SCM debugging pilot (20 outer refits per cell)
per stage: RCT n / OBS 5000; total generated per replicate: $3n / 15000$

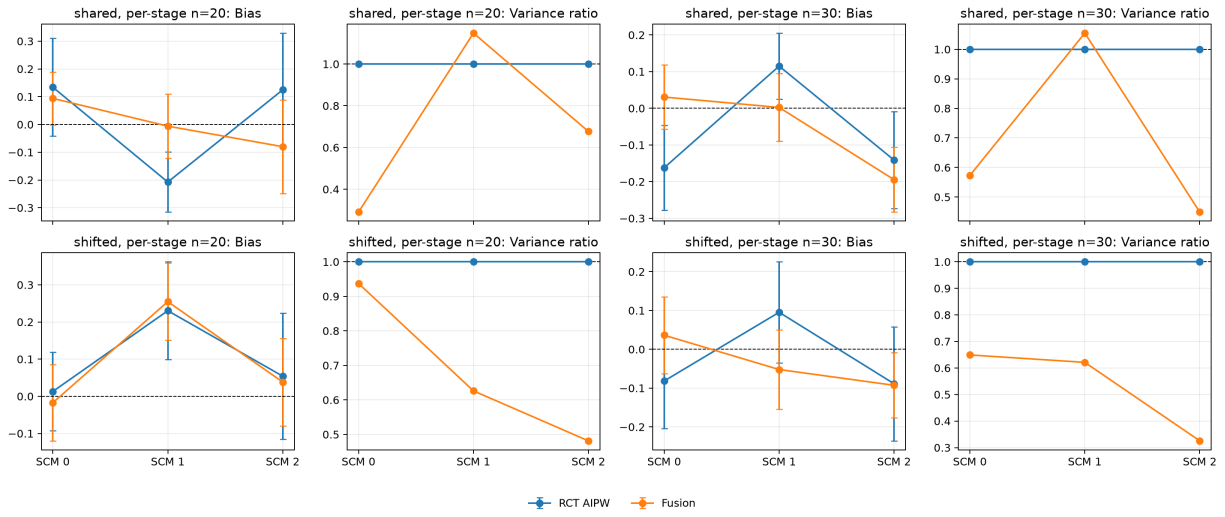


Figure 3: Earlier SSEM ATE pilot. This artifact predates the nested four-method comparison and should be interpreted only as a design and implementation check.

B. Proof Details

B.1. Conditional validity of the trial pseudo-outcome

Fix $q = (q_0, q_1)$ and condition on X . Under trial randomization and consistency,

$$\mathbb{E}_R \left[\frac{A}{e(X)} \{Y - q_1(X)\} \mid X \right] = \mathbb{E}_R \{Y(1) - q_1(X) \mid X\}, \quad (64)$$

$$\mathbb{E}_R \left[\frac{1 - A}{1 - e(X)} \{Y - q_0(X)\} \mid X \right] = \mathbb{E}_R \{Y(0) - q_0(X) \mid X\}. \quad (65)$$

Substituting these two equalities into the definition of φ cancels the two fitted regressions and yields $\mathbb{E}_R \{\varphi(V; q) \mid X\} = \tau(X)$. Applying the identity to $q = \hat{\mu}_R$, $q = \hat{\mu}_O$, and their affine blend gives

$$\mathbb{E}_R(Z_\lambda \mid X) = \tau(X), \quad \mathbb{E}_R(\Delta \mid X) = 0. \quad (66)$$

These statements are conditional on the pre-evaluation sigma-field so that the fitted functions are fixed.

B.2. Proof of Theorem 1

Using $\mathbb{E}_R(Z_\lambda) = \theta$ and exact transport,

$$\mathbb{E}_O\{r(X)g(X)\} = \mathbb{E}_R\{g(X)\}, \quad (67)$$

the conditional mean of Equation (12) is θ . Rewrite the centered estimator as

$$\hat{\theta}_r(\lambda, \omega) - \theta = (\mathbb{P}_{R,n} - P_R)\{Z_\lambda - \omega g\} + (\mathbb{P}_{O,N} - P_O)\{\omega r g\}. \quad (68)$$

The two summands are independent conditional on $\mathcal{F}$. The variance of an empirical mean of m independent observations is the population variance divided by m , which proves Equation (14). Unbiased sample variances on the two independent evaluation blocks therefore give the displayed fixed-selected-coefficient variance estimator in Section 3. Coverage of its Wald interval additionally uses the stated conditional Lindeberg and moment assumptions; variance unbiasedness alone is not a central limit theorem.

Because $g(X)$ is fixed given X ,

$$\mathbb{E}_R\{Z_\lambda - \omega g(X) \mid X\} = \tau(X) - \omega g(X), \quad \text{Var}_R\{Z_\lambda - \omega g(X) \mid X\} = \text{Var}_R(Z_\lambda \mid X).$$

Applying the law of total variance proves Equation (15).

B.3. Proof of Theorem 2

Expand the first variance in Equation (14) using $Z_\lambda = Z_0 + \lambda\Delta$. After collecting terms and multiplying by n , the coefficient-dependent part is

$$2C\lambda - 2D\omega + A\lambda^2 - 2E\lambda\omega + B\omega^2, \quad (69)$$

which is $2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta$. The representation

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H}_{\text{ATE}} \begin{pmatrix} u \\ v \end{pmatrix} = \text{Var}_R(u\Delta - vg) + \frac{n}{N} v^2 \text{Var}_O(rg) \quad (70)$$

proves positive semidefiniteness. If $AB - E^2 > 0$, the first-order condition $\mathbf{H}_{\text{ATE}} \beta = -\ell$ has the unique solution

$$\lambda^\star = \frac{ED - BC}{AB - E^2}, \quad \omega^\star = \frac{AD - EC}{AB - E^2}. \quad (71)$$

When $\mathbf{H}_{\text{ATE}}$ is singular, nonnegativity of the variance quadratic implies $\ell \in \text{Range}(\mathbf{H}_{\text{ATE}})$; otherwise a null-space direction would make the expression arbitrarily negative. The complete minimizer set is therefore $-\mathbf{H}_{\text{ATE}}^+ \ell + \text{Null}(\mathbf{H}_{\text{ATE}})$. Continuity of the quadratic proves attainment on every prespecified compact feasible set.

For completeness, suppose $\mathbf{H}_{\text{ATE}}$ is positive definite and write

$$q(\beta) = 2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta, \quad \beta^\star = -\mathbf{H}_{\text{ATE}}^{-1} \ell.$$

Completing the square gives

$$q(\beta) - q(\beta^\star) = (\beta - \beta^\star)^\top \mathbf{H}_{\text{ATE}} (\beta - \beta^\star).$$

The norm induced by a positive-definite matrix is equivalent to the Euclidean norm. Hence this distance is coercive and strictly convex. On every nonempty closed convex feasible set, it therefore attains a unique minimum, namely the $\mathbf{H}_{\text{ATE}}$ -metric projection of $\beta^\star$.

Finally, because g is X -measurable,

$$E = \mathbb{E}_R[\{g(X) - P_R g\} \mathbb{E}_R(\Delta \mid X)] = 0, \quad (72)$$

$$D = \text{Cov}_R\{\mathbb{E}_R(Z_0 \mid X), g(X)\} = \text{Cov}_R\{\tau(X), g(X)\}. \quad (73)$$

If $A = \text{Var}_R(\Delta) = 0$, then Δ is constant almost surely, and $\mathbb{E}_R(\Delta \mid X) = 0$ forces that constant to be zero. If

$$B = \text{Var}_R(g) + \frac{n}{N} \text{Var}_O(rg) = 0,$$

then g is constant P_R -almost surely and rg is constant P_O -almost surely. Exact transport gives $P_O(rg) = P_R(g)$, so the two constants agree and $\mathbb{P}_{O,N}(rg) - \mathbb{P}_{R,n}(g) = 0$ for every evaluation sample. Finally, when $E = 0$ and $A, B > 0$, completing the two scalar squares gives

$$q(\lambda, \omega) = A \left(\lambda + \frac{C}{A} \right)^2 + B \left(\omega - \frac{D}{B} \right)^2 - \frac{C^2}{A} - \frac{D^2}{B}.$$

Therefore minimization over a rectangular feasible set is exactly coordinatewise clipping of $-C/A$ and D/B . This argument does not apply to coupled constraints or to a nonzero interaction E . Substitution yields Equation (24). Since the population objective is minimized over a set containing $(0, 0)$, its oracle value cannot exceed the trial-only value. This comparison concerns the population objective, not coefficients estimated from a finite tuning sample.

B.4. Proof of Corollary 2

Write the conditional variance as

$$V(\beta) = \mathcal{V}_0 + \frac{1}{n} \{2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta\}, \quad \mathcal{V}_0 = \frac{1}{n} \text{Var}_R(Z_0).$$

When $AB - E^2 > 0$, the matrix $\mathbf{H}_{\text{ATE}}$ is positive definite and the unique minimizer is $\beta^* = -\mathbf{H}_{\text{ATE}}^{-1} \ell$. Substitution gives

$$V(\beta^*) = \mathcal{V}_0 - \frac{1}{n} \ell^\top \mathbf{H}_{\text{ATE}}^{-1} \ell.$$

For $\mathbf{H}_{\text{ATE}} = \begin{pmatrix} A & -E \\ -E & B \end{pmatrix}$,

$$\begin{aligned} \mathbf{H}_{\text{ATE}}^{-1} &= \frac{1}{AB - E^2} \begin{pmatrix} B & E \\ E & A \end{pmatrix}, \\ \ell^\top \mathbf{H}_{\text{ATE}}^{-1} \ell &= \frac{BC^2 - 2ECD + AD^2}{AB - E^2}. \end{aligned}$$

Positive definiteness makes this quadratic form nonnegative and makes it zero exactly when $\ell = 0$. Setting $E = 0$ yields $C^2/A + D^2/B$.

If $\mathbf{H}_{\text{ATE}}$ is singular, boundedness below of the variance objective implies $\ell \in \text{Range}(\mathbf{H}_{\text{ATE}})$. Evaluating at the minimum-norm solution $-\mathbf{H}_{\text{ATE}}^+ \ell$ gives

$$V_{\min} = \mathcal{V}_0 - \frac{1}{n} \ell^\top \mathbf{H}_{\text{ATE}}^+ \ell.$$

The pseudoinverse is positive definite on $\text{Range}(\mathbf{H}_{\text{ATE}})$, so the reduction is strictly positive exactly when $\ell \neq 0$. Finally, continuity gives attainment on every nonempty compact feasible set $\mathcal{B}_\beta$. If $0 \in \mathcal{B}_\beta$, the constrained minimum is at most $V(0) = \mathcal{V}_0$, and it is strictly smaller exactly when some feasible β makes $2\ell^\top \beta + \beta^\top \mathbf{H}_{\text{ATE}} \beta < 0$.

B.5. Prediction-powered CATE risk identity

Fix t and condition on X . Since $\mathbb{E}_R(Z_\lambda - \tau(X) \mid X) = 0$,

$$\mathbb{E}_R\{Z_\lambda - t(X)\}^2 = \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2 + \mathbb{E}_R\{t(X) - \tau(X)\}^2. \quad (74)$$

Exact transport also gives

$$\mathbb{E}_O\{r(X)(g(X) - t(X))^2\} = \mathbb{E}_R\{(g(X) - t(X))^2\}. \quad (75)$$

The correction in Equation (31) therefore has mean zero, which proves Equation (33). In particular, population expected loss is flat in ω . The part depending on λ is

$$Q(\lambda) = Q(0) + 2C\lambda + A\lambda^2, \quad (76)$$

so when $A > 0$ its minimizer is $-C/A$, the same integrated-noise oracle as in the population-orthogonal ATE problem. If $A = 0$, then $\Delta = 0$ almost surely by conditional mean zero, and Q is flat in λ .

For a fixed t and λ , the empirical loss is the sum of independent empirical means of $U_{t,\lambda} - \omega G_t$ under the trial and $\omega r G_t$ under the observational law. Expanding its variance as a quadratic in ω and differentiating gives Equation (35). Because t is fixed in this calculation, the formula does not identify the coefficient minimizing the risk of a learner chosen after seeing the same loss.

For the linear sieve, expand Equation (31) in β , add $\rho\beta^\top P\beta$, and set the gradient to zero. The resulting normal equation is

$$\{(1 - \omega)G_R + \omega G_{O,r} + \rho P\}\beta = h_\lambda + \omega d_g, \quad (77)$$

which proves Equation (57) whenever the matrix is invertible.

B.6. Pointwise noise coefficients and square completion

For Proposition 5, measurability and conditional mean zero give

$$\mathbb{E}_R\{Z_0 + \lambda(X)\Delta \mid X\} = \tau(X) + \lambda(X)\mathbb{E}_R(\Delta \mid X) = \tau(X).$$

At a fixed x , expand the conditional noise as a quadratic in a scalar u :

$$Q_x(u) = Q_x(0) + 2u \text{Cov}_R(Z_0, \Delta \mid X = x) + u^2 \text{Var}_R(\Delta \mid X = x).$$

A positive quadratic coefficient yields the displayed pointwise ratio. If it is zero, $\Delta = 0$ conditionally almost surely and every u -dependent term vanishes.

For Proposition 9, expand the trial contribution as

$$\{Z_\lambda - t\}^2 - \omega\{g - t\}^2 = (1 - \omega)t^2 - 2(Z_\lambda - \omega g)t + Z_\lambda^2 - \omega g^2.$$

Since $(1 - \omega)\tilde{Z}_{\lambda,\omega} = Z_\lambda - \omega g$, completing the square leaves

$$Z_\lambda^2 - \omega g^2 - (1 - \omega)\tilde{Z}_{\lambda,\omega}^2 = -\frac{\omega}{1 - \omega}(Z_\lambda - g)^2.$$

Averaging and adding the transported OBS term proves the proposition. For a linear sieve, the Hessian of the original t -dependent quadratic is a nonnegative weighted sum of G_R and $G_{O,r}$ for $0 \leq \omega \leq 1$, which proves positive semidefiniteness in that range. The claim does not extend automatically to nonlinear parameterizations.

B.7. Loss-targeted balance and approximate drift

The two inequalities in Proposition 7 follow directly from the prespecified inclusion $\mathfrak{C}_R \subseteq \mathfrak{C}$. Strict inclusion alone does not imply a strict risk inequality.

For Proposition 8, condition on $\mathcal{F}_{\text{train}}$. Expanding the shared-reference score gives

$$\mathbb{E}\{\widehat{\mathcal{R}}_c^{\text{ref}} \mid \mathcal{F}_{\text{train}}\} = Q(\lambda_{\text{ref}}) + \mathcal{R}_{\mathcal{D}_{\text{tr}}}(c) + \delta_{r,c} + \delta_{\text{nuis},c}.$$

Compare the empirical score of $\widehat{c}$ with that of an arbitrary $c \in \mathfrak{C}$. Two validation deviations contribute at most $2\varepsilon_{\text{val}}$, and the difference between two ratio and nuisance drifts contributes at most $2\Delta_r + 2\Delta_{\text{nuis}}$. Taking the infimum gives the stated oracle inequality.

Suppose a nonnegative out-of-sample-evaluable weight w satisfies

$$P_O\{w(g-t)^2\} = P_R\{(g-t)^2\} \quad \text{for every } t \in \mathcal{T}. \quad (78)$$

Replacing r by w then leaves the population correction exactly zero for every candidate, so the target identity remains valid. If $t_\beta = b^\top \beta$, expanding

$$(g - b^\top \beta)^2 = g^2 - 2\beta^\top b g + \beta^\top b b^\top \beta \quad (79)$$

shows that balancing $b g$ and $b b^\top$ preserves the minimizer. Balancing g^2 is needed only to preserve the absolute loss value.

For approximate weights, the difference between the weighted and target population losses is bounded uniformly over $\mathcal{T}$ by $|\omega| \Delta_{\mathcal{T}}(w)$. Let t_w minimize the weighted population loss and t_0 minimize the target loss. Applying the uniform bound at both minimizers gives

$$L_0(t_w) \leq L_w(t_w) + |\omega| \Delta_{\mathcal{T}}(w) \quad (80)$$

$$\leq L_w(t_0) + |\omega| \Delta_{\mathcal{T}}(w) \quad (81)$$

$$\leq L_0(t_0) + 2|\omega| \Delta_{\mathcal{T}}(w), \quad (82)$$

which is the claimed excess-risk bound. If the same finite sample is used both to impose exact sieve balance and to fit β , the balanced linear and quadratic coefficient terms cancel from the empirical correction (and the constant cancels as well if g^2 is balanced). This algebra explains why weight learning and candidate fitting require honest separation or a separate outer-fold argument.

B.8. A two-stratum oracle calculation

Let $P_R(X=0) = P_R(X=1) = 1/2$, $\tau(0) = 0$, $\tau(1) = 2$, and $g = \tau$. With $n = 100$, $N = 900$, and a common covariate marginal,

$$\mathbb{E}_R\{\tau(X)\} = 1, \quad \text{Var}_R\{\tau(X)\} = 1, \quad \omega_{\text{ATE}}^* = \frac{1}{1 + 100/900} = 0.9.$$

The baseline between- X variance is 0.01, whereas the oracle component is

$$\frac{\text{Var}_R\{0.1\tau(X)\}}{100} + \frac{0.9^2 \text{Var}_R\{\tau(X)\}}{900} = 0.0001 + 0.0009 = 0.001.$$

The reduction 0.009 concerns only between- X variation; total variance also contains the within- X pseudo-outcome noise from Equation (15).

B.9. Honest variance estimation and conditional Wald inference

For Proposition 2, condition on $\mathcal{F}$. The selected coefficients and fitted scores are fixed. Ordinary unbiased sample-variance identities give

$$\mathbb{E}\{s_{R,\text{ev}}^2(\psi_R) \mid \mathcal{F}\} = \text{Var}_R(\psi_R), \quad \mathbb{E}\{s_{O,\text{ev}}^2(\psi_O) \mid \mathcal{F}\} = \text{Var}_O(\psi_O).$$

Dividing by n and N , then adding, proves conditional unbiasedness of $\hat{V}$ by independence of the two final blocks.

For Theorem 3, center the two final-score arrays at their conditional means. Their weighted sum is $\hat{\theta}_r - \theta$. The conditional Lindeberg condition and nondegenerate variance limit give the two-sample triangular-array central limit theorem. Under a uniform conditional $(2 + \delta)$ -moment bound, the RCT Lyapunov remainder is of order $n^{-\delta/2}$, and the OBS remainder also vanishes when $n/N \rightarrow \kappa < \infty$. The same moment control gives ratio consistency of the combined sample-variance estimator. Slutsky's theorem yields the studentized limit and Wald coverage.

B.10. Estimated density ratios and outer-fold honesty

For an independently trained $\hat{r}$, subtract and add $P_O(rg) = P_R(g)$ to obtain the exact conditional drift

$$\mathbb{E}\{\hat{\theta}_{\hat{r}}(\lambda, \omega) \mid \mathcal{F}\} - \theta = \omega P_O[\{\hat{r}(X) - r(X)\}g(X)].$$

With $h_n = (\hat{r} - r)g$, the conditional second moment of the OBS evaluation perturbation is

$$\mathbb{E}\left[\left\{\sqrt{n}\omega(\mathbb{P}_{O,N} - P_O)h_n\right\}^2 \mid \mathcal{F}\right] = \omega^2 \frac{n}{N} \|h_n - P_O h_n\|_{L_2(P_O)}^2.$$

This proves the two negligible-drift requirements in Section 3; a nonnegligible ratio-training influence function or empirical remainder must instead enter the limiting influence function and its covariance.

For Proposition 3, condition on the prespecified or independently randomized split assignments and on one outer complement. Every fitted object is fixed, and the held-out paired folds retain the required RCT and OBS laws. The exact centering argument applies within that fold. Iterated expectation proves aggregate unbiasedness. It does not make different fold statistics independent: in general,

$$\text{Var}\left(\sum_k a_k \hat{\theta}_k\right) = \sum_k a_k^2 \text{Var}(\hat{\theta}_k) + 2 \sum_{k < \ell} a_k a_\ell \text{Cov}(\hat{\theta}_k, \hat{\theta}_\ell).$$

The second sum need not vanish because outer training complements overlap.

B.11. Counterexamples to adaptive reuse

The fixed-coefficient theorem cannot be extended by selecting a coefficient on the same final outcomes without a separate analysis.

Reuse through λ . Let X be degenerate, $\theta = 0$, $Z_0 = 0$, and let the Δ_i be i.i.d. Rademacher variables. Every fixed λ satisfies $\mathbb{E}\{\lambda \mathbb{P}_{R,n} \Delta\} = 0$. If the same outcomes select $\hat{\lambda} = \text{sign}(\mathbb{P}_{R,n} \Delta)$, then

$$\mathbb{P}_{R,n}(Z_0 + \hat{\lambda} \Delta) = |\mathbb{P}_{R,n} \Delta|,$$

whose expectation is strictly positive.

Reuse through ω . Under the common covariate marginal, let $g(X)$ be a nondegenerate mean-zero Rademacher variable and let the RCT-valid score be identically zero. Every fixed ω centers $\omega\{\mathbb{P}_{O,N}g - \mathbb{P}_{R,n}g\}$. Choosing $\hat{\omega}$ as the sign of that same bracket produces its absolute value, which again has strictly positive expectation. The bias in both examples comes solely from adaptive reuse.